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Analysis of High-Speed Rail Passenger Travel Behaviour Considering Passenger Heterogeneity Under Dynamic Pricing Strategy

Siye Guo¹  | Yun Jing^{2,3}  | Siyuan Wang²  | Hengzhi Liu⁴  | Yiao Liu² 
¹China Academy of Transportation Sciences, Beijing, China | ²School of Traffic and Transportation, Beijing Jiaotong University, Beijing, China | ³Intelligent High Speed Rail System Frontier Science Center, Beijing Jiaotong University, Beijing, China | ⁴School of Electrical Engineering, Beijing Jiaotong University, Beijing, China

Correspondence: Yun Jing (yjing@bjtu.edu.cn)

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ABSTRACT

Clarifying passengers' travel behaviour under dynamic pricing strategies for high-speed railway (HSR) can provide theoretical foundations and practical insights for railway operators to develop personalized and differentiated marketing strategies and pricing schemes. Firstly, this paper extends the theory of planned behaviour (TPB) by introducing 'travel habits' and 'perceived travel fairness' as latent variables, constructing a psychological decision-making framework that accounts for passenger heterogeneity under HSR dynamic pricing strategies. Secondly, employing revealed preference (RP) and stated preference (SP) survey methods combined with D-optimal experimental design, we designed a realistic experimental scenario to accurately capture passenger choice behaviour under dynamic pricing conditions. Then, by integrating the multiple indicators multiple causes (MIMIC) model and latent class model (LCM), we developed a hybrid model to analyse passenger heterogeneity from both latent variable and latent class perspectives, uncovering behavioural patterns and segmenting passenger groups. Finally, empirical analysis based on survey data revealed that the hybrid model classified passengers into four distinct segments: Time-sensitive (24.37%), cost-sensitive (20.03%), scenario-dependent (41.27%) and experience-oriented (14.33%). The model systematically elucidates the behaviour habits and preference characteristics of each passenger segment, providing robust support for the sustainable reform and optimization of HSR dynamic pricing strategies.

1 | Introduction

As a critical component of China's modern, high-quality integrated transportation network, high-speed rail (HSR) effectively meets passengers' demand for premium travel services, significantly reduces intercity travel time and fosters regional economic development. With the continuous advancement of national strategies and the rapid expansion of railway infrastructure, China's HSR development has transitioned from a 'large-scale construction' phase to a 'large-scale operation' phase, shifting its primary focus from infrastructure development to transportation

management and service optimization. Fare pricing serves as a crucial link between supply and demand, functioning as an economic lever to optimize ticket pricing systems, improve seat allocation efficiency, and rationally guide passenger flow. China's railway pricing policy has evolved through four key stages: strict fare regulation, deregulation under centralized management, diversified adjustments through price hearings, and the implementation of market-based pricing. In accordance with the Notice on Reforming and Improving the Passenger Fare Policy for High-Speed EMU Trains (issued by the National Development and Reform Commission on 19 February 2016), China State

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Railway Group Co., Ltd. (previously China Railway Corporation) is authorized to implement dynamic fare discounts based on market competition conditions and passenger flow distribution patterns [1].

In time-varying transportation markets, heterogeneous passengers—characterized by distinct individual attributes and travel behaviour patterns—exhibit complex, multidimensional demand characteristics. Establishing the mapping relationship between market segmentation and price elasticity not only preserves the essential public service nature of railway operations but also facilitates pareto-optimal improvements in resource allocation. Conventional single-dimensional segmentation approaches fail to fully exploit market potential, thereby constraining the effectiveness of pricing strategies and resource allocation in revenue management systems, ultimately compromising overall corporate profitability. Scientific passenger segmentation reveals the heterogeneity in choice preferences among different traveller segments under high-speed rail dynamic pricing. This insight provides a theoretical foundation for railway operators to develop personalized and differentiated marketing strategies and pricing schemes based on passenger value tiers. This approach serves as a crucial reference for transforming railway operations from traditional marketing models to a win-win strategy that maximizes both corporate profits and service quality, with passenger value at its core.

Guided by passenger market demand, this paper explores innovative methods that intersect behaviouralistics and social psychology. It designs a psychometric decision-making framework for high-speed rail travellers under dynamic pricing strategies, taking into account traveller heterogeneity. The paper constructs a discrete choice model of traveller behavioural intentions under dynamic pricing strategies, delves into the impact of heterogeneity on traveller choice, elucidates the mechanisms by which traveller heterogeneity affects travel decision-making, and precisely delineates the behavioural patterns of heterogeneous travellers under dynamic pricing. The research offers significant theoretical support and parameter inputs for the design of dynamic pricing schemes in high-speed rail.

This paper is organized as follows. Section 2 summarizes the related work involved in this article. Section 3 constructs a framework for describing the psychological decision-making process of high-speed rail heterogeneous passenger travel behaviour based on the Theory of Planned Behaviour. Section 4 designs and implements a survey questionnaire on passenger travel behaviour under dynamic pricing strategies to deeply understand the patterns of passenger flow changes and explore the elasticity of heterogeneous passenger demands. Section 5 builds an integrated model, multiple indicators multiple causes-latent class model (MIMIC-LCM) based on structural equation modelling and latent class methods to analyse heterogeneous passenger choice behaviour from the perspectives of latent variables and latent categories, revealing passenger behavioural characteristics and segmenting passenger groups. Section 6 validates the model effectiveness by scenario dependence and interpretation of passenger heterogeneity from the perspectives of latent variables and latent categories through examples. Conclusion and future prospects are introduced in the last section.

2 | Literature Review

2.1 | Travel Behaviour Analysis

To examine the impact on passengers' travel choice behaviour and reveal the influencing factors, scholars worldwide have accumulated extensive research based on disaggregate choice models for travel behaviour analysis.

2.1.1 | Research on Travel Behaviour Based on Disaggregate Choice Models

Since the 1970s, McFadden et al. [2] systematically established the theoretical framework of disaggregate models (DM) based on random utility theory, laying the foundation for subsequent DM-based methodologies. Scholars worldwide have since developed a series of disaggregate models, including mixed logit (MXL), multinomial logit (MNL) and nested logit (NL) models.

However, Hensher et al. [3] argued that the MXL model lacks sufficient accuracy in capturing travel behaviour and proposed an improved semi-parametric extended MNL model, demonstrating superior predictive performance. Hu et al. [4] applied the MNL model with mobile signalling data to analyse multi-modal trip chaining behaviour in Shanghai, providing insights for developing efficient and sustainable transportation systems aligned with traveller preferences. Cheng [5] constructed an MNL model based on random utility maximization theory to examine high-speed rail passengers' choice behaviour, revealing significant influences from departure time window, ticket price and station stops.

A key limitation of the MNL model is its reliance on the IIA (independence of irrelevant alternatives) assumption, which may not hold in real-world scenarios. The NL model overcomes this constraint. For instance, Zhang et al. [6] developed an NL model incorporating travel inertia to simulate airport ground access mode choices, optimizing the trade-off between rail capacity utilization, service quality, and operational revenue. Jin et al. [7] designed a nested NL model to analyse passenger preferences under different scenarios, facilitating demand balancing between conventional and high-speed rail services for revenue enhancement.

Despite their widespread adoption, disaggregate models fail to account for latent variables (e.g., psychological factors, attitudes, perceptions) that indirectly influence travel decisions. Structural equation modelling (SEM) has thus gained traction in travel behaviour research by explicitly incorporating latent variables as exogenous/endogenous factors, enabling deeper analysis of their behavioural impacts.

2.1.2 | Research on Travel Behaviour Based on SEM

SEM, a multivariate statistical linear parameter model, has been widely applied in psychology, education and social sciences [8]. However, its adoption in transportation behaviour research emerged relatively later. Li et al. [9] employed SEM to analyse how socioeconomic attributes, psychological factors and urban transport infrastructure supply characteristics collectively

influence travel choice behaviour. Their study demonstrated SEM's capability to integrate observed and latent variables in explaining complex decision-making processes. In the context of intermodal competition, Chu et al. [10] developed an SEM framework incorporating intercity transfer behaviour and network information guidance, revealing the underlying logic behind mode choices between air and rail. This approach provided insights into traveller preference formation under competitive service environments. Ye et al. [11] further expanded SEM applications to shared mobility systems, constructing a model to identify key factors affecting car-sharing usage intention and parking choice behaviour.

2.1.3 | Heterogeneity Analysis Based on SEM

Early research on travel behaviour primarily focused on extracting behavioural characteristics and mode choice patterns, largely overlooking traveller heterogeneity. As personalized travel demands become increasingly prominent, the significant heterogeneity among traveller groups have gained widespread theoretical recognition. Consequently, analysing all travellers as a homogeneous population is no longer methodologically appropriate. This underscores the necessity of employing SEM to examine the intrinsic relationships among psychological latent variables, thereby enabling heterogeneous group behaviour analysis. Ji et al. [12] developed a nested model combining *K*-means clustering and CHAID algorithms to identify highway traveller segments, extracting segmentation rules and group characteristics across different time periods to examine how demand intensity influences behavioural heterogeneity. Similarly, Liang et al. [13] constructed an SEM framework, treating individual characteristics as exogenous variables and departure time, trip attributes and travel purpose as endogenous variables, to analyse public transit usage patterns among commuters with high, medium and low trip stability.

It can be seen that most of the existing research on travellers' travel choice behaviour focuses on the impact of directly observable dominant variables, such as individual socio-economic attributes, travel characteristics and alternative attributes on travellers' travel choice behaviour, while ignoring the psychological factors such as travellers' perceptions, attitudes and cognitive effects will have an important impact on their travel mode decisions.

2.2 | Individual Heterogeneity

Travellers, as behavioural agents with distinct individual attributes and travel characteristics, exhibit heterogeneity that significantly influences their decision-making outcomes. Effectively analysing such heterogeneity is crucial for transportation enterprises, as it directly impacts pricing strategies and resource allocation in revenue management, ultimately affecting overall corporate profitability. The concept of heterogeneity originates from Adam Smith's 'economic man' hypothesis in Western economics, which posits that individuals make decisions to maximize their own interests [14]. In the field of transportation, scholars have primarily explored

individual heterogeneity by examining its attribute sources and measurement methods.

2.2.1 | Sources of Heterogeneity Attributes

The choice of travel mode is primarily influenced by four categories of factors: travellers' socioeconomic attributes, trip characteristics, mode-specific attributes, and psychological decision variables. Traveller heterogeneity manifests predominantly in socioeconomic characteristics and psychological decision-making variables. Frédéric et al. [15] identified key drivers of social inequality in high-speed rail travel by examining the socio-economic characteristics of passengers. Chung et al. [16] investigated the impact of high-speed rail on passenger behaviour by studying passenger characteristics and their satisfaction with rail services. Wang et al. [17] incorporated buyers' fluctuating psychological valuations during the advance selling period into a presale model constrained by seller capacity, considering consumer heterogeneity and demand uncertainty.

2.2.2 | Passenger Market Segmentation

Measuring traveller heterogeneity is fundamentally an exercise in market segmentation within the passenger transport sector. By leveraging differences in travellers' natural attributes and demand preferences—while accounting for both diverse consumer needs and the limited resources of transport enterprises—this approach categorizes travellers into distinct groups. Such segmentation enables targeted marketing strategies, facilitating the optimization of transport products and enhancing corporate revenue.

The most commonly used methods for measuring passenger heterogeneity are the random parameters logit model [18, 19] and LCM [20, 21]. Monsuur et al. [18] employed a latent class ordered logit model to segment passenger groups, emphasizing the importance of unobserved heterogeneity in statistical data. Peng et al. [19] used the MNL model to reveal how individual and travel attributes quantitatively influence the sensitivity of route choice factors. Compared to the random parameter's logit model and traditional clustering methods, the latent class model offers advantages in capturing behavioural heterogeneity. It can identify potential latent variables without excluding other variables, retain manifest variables completely, allow cross-processing of any variable type, and enhance the interpretability of classification results. Zhang et al. [20] applied the LCM model to reveal choice heterogeneity across different passenger groups and identify factors influencing pre-trip reserve time choices, thereby optimizing the integration between high-speed rail hubs and urban transport networks. Li et al. [21] developed an LCM model incorporating passengers' socio-economic characteristics, travel features, and coach service frequency to identify road passenger travel behaviour patterns and analyse their characteristics, guiding intercity coach operators in designing precise ticket distribution services for specific passenger types.

Research on heterogeneity in the domestic transportation field is relatively limited, primarily focusing on how passenger heterogeneity influences travel choice behaviour [22, 23] and aids transportation enterprises in optimizing operational plans and

marketing strategies [24, 25]. Based on studies of individual differences and factors affecting passenger choice behaviour, Su et al. [22] used latent class and random parameter logit models to explore the impact of passenger attributes and train characteristics on choice heterogeneity at both group and individual levels. Ren [23] segmented passengers using a latent class model to examine differences in their preferences for the economy, comfort, and convenience of transportation modes. Yang [24] adopted a Q-learning reinforcement learning algorithm to design a personalized recommendation method for intermodal bus-rail travel plans based on passenger profiles, effectively reducing travel time and costs while providing personalized services. Zhai [25] constructed high-speed rail revenue management models with and without consideration of individual heterogeneity to study its impact on revenue management.

2.3 | Dynamic Pricing

The primary objective of dynamic pricing is to maximize the revenue of railway operators under the condition of meeting passenger travel demands, based on the travel behaviours of different types of passengers. Russo et al. [26] designed a dynamic high-speed rail pricing model that incorporates reference time variables to identify and compare how ticket prices vary with passenger behaviour and time. Martin et al. [27] proposed a multi-agent reinforcement learning framework based on non-homogeneous Markov games, combined with random utility models to capture passenger behavioural decision-making, thereby designing dynamic pricing strategies in the context of operator cooperation and competition. Chen et al. [28] utilized gradient boosting decision trees to segment the feature space of passenger travel choices for price stratification, constructing a pricing model aimed at maximizing railway operator revenue and increasing daily earnings. Kaan [29] analysed passenger travel behaviour based on factors such as departure time and travel purpose, establishing a dynamic pricing model grounded in passenger behaviour.

As a primary mode for medium- to long-distance travel, high-speed rail should integrate traveller heterogeneity into demand analysis to identify behavioural patterns. These insights serve as critical inputs for dynamic modelling, forming the foundation for railway operators to implement revenue management strategies—including service optimization, ticket pricing policies, and capacity allocation.

2.4 | Review of Research Status

A comprehensive comparison of existing studies is summarized in Table 1.

In summary, existing research on travel behaviour analysis lacks comprehensive consideration of the heterogeneous characteristics of high-speed rail passengers and has yet to establish a robust theoretical framework for heterogeneity. The applied analytical models remain limited, predominantly relying on mixed logit models to address heterogeneity—while flexible, these fail to explicitly classify latent passenger segments. Moreover, no integrated behavioural theory framework has been developed to systematically examine the psychological factors influencing

passengers' choice behaviour under dynamic pricing strategies or to explore interdependencies among these factors. This gap contributes to insufficient clarity among industry practitioners regarding the coupling mechanisms between fare structures and travel decision-making.

To facilitate the transition of railway operations from traditional marketing models to a win-win strategy that maximizes railway benefits centred on passenger value and optimizes transportation service quality, this paper makes the following main contributions:

1. Based on the theory of planned behaviour (TPB), we introduce expanded latent variables—‘Travel Habits’ and ‘Perceived Travel Fairness’—to construct a quantitative psychological decision-making framework for high-speed rail passengers under dynamic pricing strategies. This expanded TPB-based framework further enhances model interpretability.
2. A comprehensive survey was conducted to examine the choice behaviour of heterogeneous passenger groups under dynamic pricing. Using a D-optimal experimental design, highly realistic service scenarios were constructed to accurately capture passenger decision-making in response to dynamic fare changes.
3. Accounting for both preference heterogeneity and response heterogeneity among travellers, an integrated MIMIC-LCM model was developed by combining structural equation modelling with latent class methods. This approach enables a dual examination of passenger behaviour from the perspective of latent variables and latent classes, revealing behavioural patterns and supporting passenger segmentation.
4. Model calibration based on behavioural questionnaire data confirmed that the proposed approach achieves high goodness-of-fit. The model effectively captures and interprets the behavioural mechanisms of heterogeneous passengers under dynamic pricing, thereby contributing to the theoretical advancement of travel choice behaviour modelling.

3 | Quantitative Framework of Travel Psychological Decision-Making of Heterogeneous High-Speed Rail Passengers

3.1 | Extended Variables of TPB

The TPB, as a universal foundational theoretical model, when applied to specific research subjects in a particular field, may have certain limitations if used singularly due to specific factors that may influence behavioural intentions in the specific study. Considering the heterogeneity in choice behaviour among high-speed rail passengers when faced with multiple train options, this study extends the original TPB by introducing two additional constructs: ‘Travel Habits’ and ‘Perceived Travel Fairness’. This approach constructs a psychological decision-making framework and model structure for understanding the travel behaviour of heterogeneous high-speed rail passengers, thereby further enhancing the model's interpretability.

TABLE 1 | Comparison of existing studies.

Research direction	Research method	Pros and Cons	References
Travel behaviour	Disaggregate model	Accumulate the richest travel behaviour research methods; Ignore the impact of potential variables that cannot be directly observed on travel behaviour	[2–7]
	Structural equation model	Directly observable latent variable; Model setting and validation are complicated	[8–11]
	Consider heterogeneity	Identify behavioural differences among different groups; High data quality requirements	[12, 13]
Individual heterogeneity	Source analysis	Most of them focus on explicit variables, and the capture of deep psychological motivation is limited	[14–17]
	Market segmentation	The calculation of the random coefficient logit is complex; LCM has high interpretability and is more suitable for the analysis of behavioural heterogeneity	[18–21]
	Application in travel behaviour	The domestic related research is still less; Focus on behaviour preference analysis, and its application in combination with dynamic pricing needs to be deepened	[22–25]
Dynamic pricing	Behaviour orientation	Passenger heterogeneity needs to be carefully depicted; The adaptability of the dynamic pricing application to a complex market environment needs to be verified	[26–29]

3.1.1 | Travel Habits

When passengers accumulate a certain amount of high-speed rail travel experience, they are likely to choose travel options based on habitual choices in specific decision-making environments. Habitual choices can enhance the effectiveness of ‘behavioural habits’ on ‘actual behaviour’ and have a negative feedback effect on the reaction time of ‘behavioural intentions’. However, when passengers are in a stable decision-making environment, the decision-making effects of ‘behavioural habits’ and ‘behavioural intentions’ on actual choice behaviour remain consistent. In many related empirical studies, it is known that the expanded variable of ‘travel habits’ has a significant relationship with the original variables of the TPB theory [30–32].

Unlike passengers of other transport modes, high-speed rail passengers exhibit distinct characteristics—prioritizing comfort, safety, punctuality, and the system’s quasi-public service attributes—which drive consistently rising modal shares. However, under dynamic pricing strategies, travel frequency varies significantly due to passenger heterogeneity. To address this, we integrate the ‘travel habits’ variable into the TPB framework, augmenting its core constructs to unravel the decision-making mechanisms of heterogeneous high-speed rail passengers in response to dynamic pricing.

3.1.2 | Perceived Travel Fairness

Passengers’ decisions to choose high-speed rail are significantly influenced by their perceived travel experience, which involves both intra-personal comparisons (against their own historical experiences) and inter-personal comparisons (relative

to other travellers). These perceptions critically shape travel propensity and provide actionable insights for railway operators to cultivate high-speed rail frequent passengers [33, 34]. This study defines such experiential evaluations as ‘perceived travel fairness’—a multidimensional construct encompassing rule fairness, opportunity fairness, and outcome fairness throughout the journey.

Given the public utility nature of high-speed rail, equitable fare rule-making strategies tailored to passengers’ socioeconomic attributes are fundamental to maximizing perceived travel fairness. Opportunity fairness in high-speed rail requires that all passengers can secure tickets for their utility-maximizing seat class prior to departure. This opportunity fairness is predicated on rule fairness—equitable pricing policies that guarantee equal access to travel options. Meanwhile, outcome fairness reflects post-trip satisfaction with service attributes such as comfort, safety, and efficiency, completing the fairness triad that shapes overall travel experience evaluations.

In summary, considering the different dimensions of perceived travel fairness for high-speed railway passengers, a summary is presented in Table 2.

3.2 | Quantitative Framework of Psychological Decision-Making

The process through which high-speed rail passengers select and complete a specific travel option represents the outcome of their psychological decision-making. Based on the theoretical foundation of the TPB, the influencing factors and their functional mechanisms throughout the entire process of high-speed rail

TABLE 2 | Dimensions of ‘perceived travel fairness’ among high-speed railway passengers.

Perception dimension of travel fairness		Definition
Rule fairness	Fair pricing	Whether the ticket price of the train they take meets their personal economic expectations
Opportunity fairness	Fair travel opportunities	Whether the selected train seat resources ensure that passengers can choose their ideal seat
Result fairness	Comfort level	The selected train’s ability to reduce travel fatigue and the duration of the journey
	Safety level	The safety performance of the selected train
	Speed level	The selected train’s ability to eliminate jet lag
	Value retention	The services obtained justify the time and effort invested

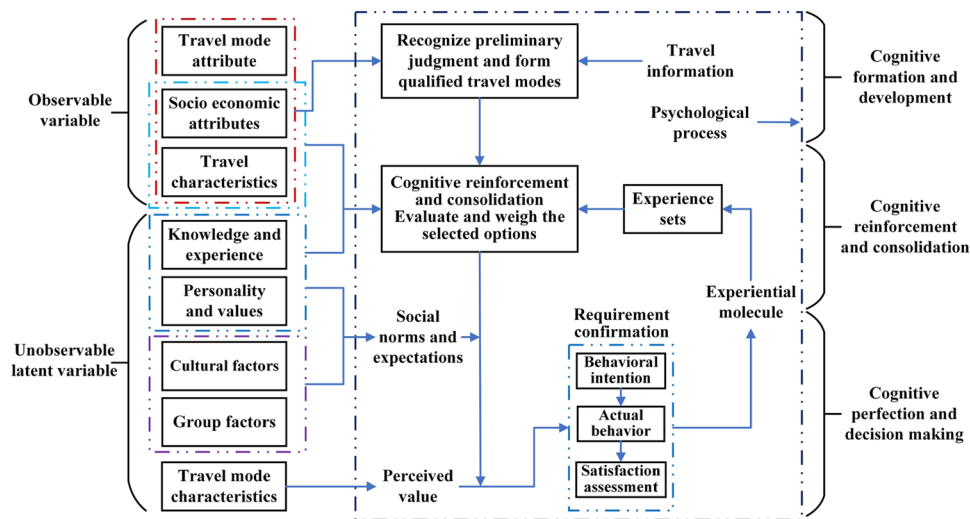


FIGURE 1 | Decision process of high-speed rail passenger travel behaviour based on TPB framework.

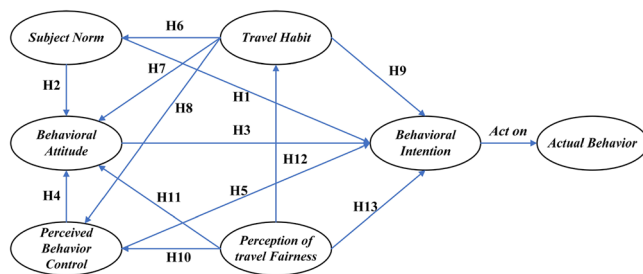


FIGURE 2 | A framework for describing the psychological decision-making process of heterogeneous high-speed rail passengers under dynamic pricing strategies.

passengers’ travel behaviour can be summarized as illustrated in Figure 1.

Incorporating the extended variables from Section 3.1, this study proposes a framework describing the psychological decision-making process of heterogeneous high-speed rail passengers under dynamic pricing strategies. The model structure is illustrated in Figure 2, with the symbols of each psychological latent variable provided in Table 3.

TABLE 3 | Explanation of symbols for various psychological latent variables.

Name of psychological latent variable	Symbol
Subject norm	SN
Behavioural attitude	BA
Perceived behaviour control	PBC
Travel habit	TH
Perceived travel fairness (PF)	Rule fairness RF
	Opportunity fairness OF
	Fairness of result FR
Behavioural intention	BI

Based on the theoretical model diagram and supported by experience and theory [9–13], this paper makes the following basic assumptions about the model: Hypothesis 1 (H1): SN of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BI. Hypothesis 2 (H2): SN of heterogeneous passengers under dynamic pricing strategies has

a significant positive effect on BA. Hypothesis 3 (H3): BA of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BI. Hypothesis 4 (H4): PBC of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BA. Hypothesis 5 (H5): PBC of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BI. Hypothesis 6 (H6): TH of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on SN. Hypothesis 7 (H7): The TH of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BA. Hypothesis 8 (H8): TH of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on PBC. Hypothesis 9 (H9): TH of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BI. Hypothesis 10 (H10): PF of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on PBC. Hypothesis 11 (H11): PF of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on BA. Hypothesis 12 (H12): PF of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on TB. Hypothesis 13 (H13): Perceived travel fairness of heterogeneous passengers under dynamic pricing strategies has a significant positive effect on behavioural intention.

4 | Design and Implementation of Passenger Travel Behaviour Questionnaire

4.1 | Survey Methods and Questionnaire Design

4.1.1 | Research Methodology

The survey methods for travel behaviour are divided into RP survey, SP survey, and RP&SP survey [35]. RP survey investigates actual actions or choices already made by individuals, while SP survey examines intended choices and decision-making considerations under hypothetical scenarios. RP survey can compensate for the lack of reliability in SP survey data, while SP survey can enhance the limitations of RP survey in selecting unknown scenarios.

Therefore, based on the strengths and weaknesses of RP and SP surveys, this study adopts the RP&SP survey method to investigate the travel behaviour of heterogeneous passengers under dynamic pricing strategies. This integrated method enables the collection of both intended choices under dynamic pricing conditions and actual travel behaviours under existing ticketing systems.

4.1.2 | Questionnaire Design

The questionnaire design for investigating the travel behaviour of high-speed rail passengers under dynamic pricing strategies is divided into five parts. Part 1 focuses on passenger travel characteristics, including departure time, preferred number of train cars, travel purpose, and days of advance ticket purchase. Part 2 covers the survey of high-speed rail passengers' travel mode choices under dynamic pricing strategies. Part 3 examines the differences in passengers' perceptions of travel time and

TABLE 4 | SP survey variables and level settings.

Variable	Horizontally disposed
Departure time	−6 h, −4 h, −2 h, 0, +2 h, +4 h, +6 h
Travel time	4 h 30 min, 5 h 30 min, 6 h 30 min
Train occupancy period	High occupancy period, non-high occupancy period
Service level	Highest, higher, commonly, lower, lowest
Ticket price	−20%, −10%, 0, +10%, +20%

Note: The '+' and '−' signs for departure time represent the increase and decrease ratios for passengers relative to their intended departure time, indicating delays and advances in departure. The '+' and '−' signs for ticket prices represent the increase and decrease ratios relative to static prices under the dynamic pricing strategy.

costs. Part 4 assesses passengers' psychological factors. Part 5 includes passengers' socio-economic attributes, such as gender, age, education level, and income level.

Parts 1 and 5 fall within the scope of the RP survey and are easily understood, so they are not elaborated on in this paper. This study primarily focuses on elaborating on the content of the SP survey in Parts 2, 3, and 4.

1. SP survey: High-speed rail passengers' travel behaviour intention under a dynamic pricing strategy

The questionnaire is based on a dynamic pricing strategy that changes ticket prices proportionally. It assumes the original static pricing scenario for passenger travel and creates different travel service scenarios by cross-combining attributes such as different departure times, travel durations, and ticket prices of high-speed trains. This is done to capture passengers' intentions regarding travel choices under dynamic pricing strategies.

Table 4 shows the variables and their levels set in the design of the scenarios for the SP survey on high-speed rail passengers' travel behaviour intention under dynamic pricing strategies.

After determining the attributes and levels of variables, 1050 different experimental scenarios were generated by randomly combining them. In order to ensure that the selected attribute level setting combination scheme can reflect the trade-off information of passengers' choice of travel scheme under the dynamic pricing strategy to the greatest extent, which is conducive to maximizing the choice preference information. This paper is based on the D-optimal experimental design method, and uses the R language for programming to eliminate the super idealized scenes with complete advantages or disadvantages. The solution process is shown in Figure 3.

Using passengers' choices of different train departure times as the baseline control and design groups for the experimental scenarios, unrealistic scenarios and those with obvious dominance or inferiority were eliminated from the generated combinations. A total of 92 sets of scenarios were ultimately formed, each comprising 12 scenarios. As an example, four scenarios from one of these sets are illustrated in Figure 4.

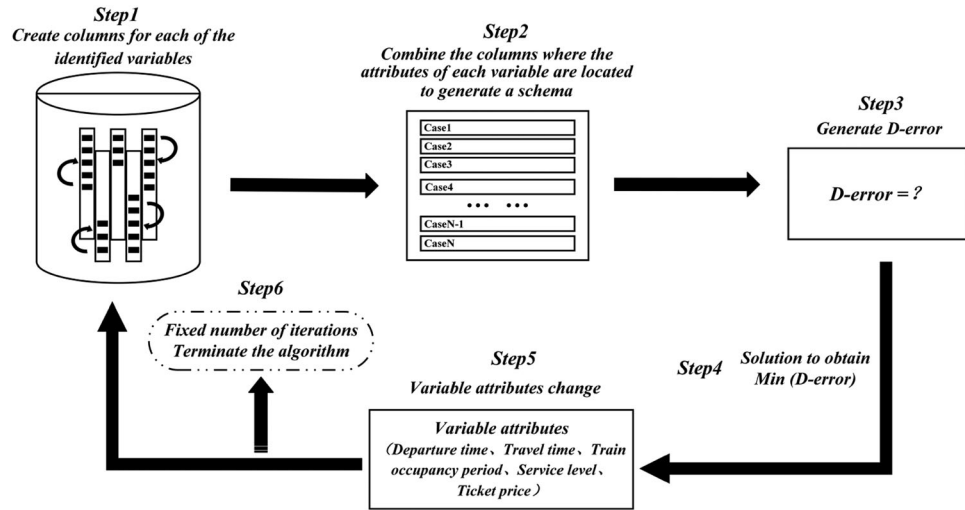


FIGURE 3 | D-optimal solving process.

Scenario 1	Train occupancy rate at departure time	Departure time	Travel time	Service level	Ticket price
Original static pricing travel plan		12:00	5h30min	Commonly	553yuan
Dynamic pricing travel plan	A	12:00	4h30min	Higher	608yuan
	B	14:00	5h30min	Commonly	498yuan
Other modes of transportation	C	Don't choose high-speed rail travel			

Scenario 2	Train occupancy rate at departure time	Departure time	Travel time	Service level	Ticket price
Original static pricing travel plan		10:00	6h30min	Commonly	553yuan
Dynamic pricing travel plan	A	10:00	4h30min	Commonly	608yuan
	B	14:00	5h30min	Lower	553yuan
Other modes of transportation	C	Don't choose high-speed rail travel			

Scenario 3	Train occupancy rate at departure time	Departure time	Travel time	Service level	Ticket price
Original static pricing travel plan		08:00	4h30min	Higher	553yuan
Dynamic pricing travel plan	A	08:00	6h30min	Lowest	442yuan
	B	10:00	5h30min	Commonly	498yuan
Other modes of transportation	C	Don't choose high-speed rail travel			

Scenario 4	Train occupancy rate at departure time	Departure time	Travel time	Service level	Ticket price
Original static pricing travel plan		06:00	5h30min	Commonly	553yuan
Dynamic pricing travel plan	A	06:00	4h30min	Highest	664yuan
	B	10:00	5h30min	Commonly	498yuan
Other modes of transportation	C	Don't choose high-speed rail travel			

FIGURE 4 | Scenario design for survey on travel behaviour intentions of high-speed rail passengers under dynamic pricing strategies.

2. Investigation and quantification of passenger transfer elasticity under a dynamic pricing strategy

Investigating and quantifying passenger transfer elasticity under dynamic pricing strategies aims to clarify demand responsiveness, providing critical parameter inputs for subsequent high-speed rail pricing models that account for traveller heterogeneity. High-speed rail passenger demand response falls under price-driven demand response, wherein travellers alter their original choice behaviour upon receiving fare adjustment signals from railway operators—either reducing or increasing travel demand for specific time slots. The primary objective of implementing dynamic pricing is to balance daily passenger flow, necessitating operators to prioritize high-demand period load shifting. This paper examines heterogeneous travellers' departure-time transfer elasticity through differentiated fare adjustment strategies. The levels of price regulation are set as follows:

Assuming you intend to travel by high-speed train during the peak occupancy period of 10:00 to 11:00, with a ticket price of 553 yuan. If the ticket price is increased during the peak occupancy period while discounted prices are offered during other periods, which travel option would you specifically choose?

Price adjustment method

- ◇ Train high occupancy rate period: increased by 5%, 581 yuan
- ◇ Other time period: 5% off, 525 yuan
- ◇ Train high occupancy rate period: increased by 10%, 608 yuan
- ◇ Other time period: 10% off, 498 yuan
- ◇ Train high occupancy rate period: increased by 15%, 636 yuan
- ◇ Other time period: 15% off, 470 yuan
- ◇ Train high occupancy rate period: increased by 20%, 664 yuan
- ◇ Other time period: 20% off, 442 yuan

Scheme selection

- ☐ Still departure from 10 to 11 o'clock
- ☐ Departure before 10 o'clock
- ☐ Departure after 11 o'clock
- ☐ Choose other transport
- ☐ Still departure from 10 to 11 o'clock
- ☐ Departure before 10 o'clock
- ☐ Departure after 11 o'clock
- ☐ Choose other transport
- ☐ Still departure from 10 to 11 o'clock
- ☐ Departure before 10 o'clock
- ☐ Departure after 11 o'clock
- ☐ Choose other transport
- ☐ Still departure from 10 to 11 o'clock
- ☐ Departure before 10 o'clock
- ☐ Departure after 11 o'clock
- ☐ Choose other transport

In this paper, the departure transfer fluctuation is introduced to express the sensitivity of the departure time selected by passengers to the change of ticket price policy. The passenger departure transfer fluctuation coefficient can be obtained as follows:

$$\tau = \frac{(T_{n+1} - T_n)/T_n}{(P_{n+1} - P_n)/P_n} \quad (1)$$

In Equation (1), τ represents the passenger departure transfer fluctuation coefficient; T_n denotes the passenger demand volume under static pricing P_n ; T_{n+1} represents the passenger demand volume after differential pricing P_{n+1} . Based on the questionnaire data and Equation (1), the obtained transfer volatility coefficient can provide key parameter input for the subsequent dynamic pricing scheme design of high-speed railway.

3. Psychological factors measurement

Based on the framework established in Section 3.2, this paper designed corresponding measurement variables to quantify the latent variables within the psychological decision-making framework of heterogeneous passengers, which cannot be directly observed. A five-point Likert scale was adopted to describe each measurement variable, with assigned values of 1 to 5 representing 'strongly disagree', 'disagree', 'neutral', 'agree', and 'strongly agree', respectively. The measurement scales for the psychological latent variables are presented in Table 5.

4.1.3 | Implementation of the Survey

Online surveys offer distinct advantages, including broad geographic coverage, user-friendly interfaces, flexible distribution methods, precise target group segmentation, and high response rates. This study employed the professional online survey platform 'Questionnaire Star Professional' to design and disseminate questionnaires, specifically targeting passengers traveling on the Beijing-Shanghai High-Speed Rail line. For the stated preference experiment design, the system randomly selected one scenario from a pre-defined library of 92 combinatorial choice sets to present to each respondent, ensuring diversified response patterns while maintaining experimental control.

The survey was conducted from 1 September to 30 September 2022, during which a total of 887 responses were collected. To ensure data quality, responses were subjected to validation checks based on time constraints, logical consistency of answers and adherence to scenario selection rules. Invalid questionnaires were excluded from the analysis, resulting in a final sample of 823 valid responses.

4.2 | Descriptive Statistical Analysis of Significant Variables

4.2.1 | Socioeconomic Attributes Analysis

By statistically analysing the socioeconomic attributes of the surveyed respondents and comparing them with all ticket data during the survey period, the statistical analysis results, as shown in Table 6, were obtained based on the behavioural survey data on dynamic pricing selection.

Based on passenger ticket data from Beijing South Station and Shanghai Hongqiao Station, the socioeconomic attributes of passengers traveling between these two stations during 30 September 30 2022, were partially extracted and compared with the survey data. Due to data availability constraints, only basic information such as age and gender could be compared; other socioeconomic attributes were not included in the analysis. The results show that the sampling accuracy exceeded 95% in terms of both gender and age distribution, indicating that the survey sample is highly representative and reliably reflects the travel behaviour of passengers under dynamic pricing strategies [35].

4.2.2 | Travel Characteristics Analysis

In terms of travel purposes, leisure activities (including tourism, shopping, and visiting relatives) and business trips constituted the majority, accounting for 40.08% and 38.06%, respectively, followed by student travellers at 12.94%. Regarding the advance ticket purchase period, passengers who purchased tickets 3–7 days before departure represented the largest proportion at 43.28%, followed by those purchasing 1–2 days in advance, accounting for 39.30%. These findings are generally consistent with the analysis of presale period distribution and evolutionary characteristics. A cross-tabulation analysis between travel purpose and advance purchase days is illustrated in Figure 5.

As shown in Figure 5, student travellers, migrant workers and those traveling for leisure purposes tend to purchase tickets within 1 week before departure, while business travellers show a stronger preference for purchasing tickets within 2 days prior to departure. Notably, 25.77% of student travellers choose to buy tickets on the very day that tickets become available, which aligns with the typical ticket purchasing psychology of students returning to school.

4.3 | Descriptive Statistical Analysis of Latent Variables

4.3.1 | Analysis of Data Reliability and Validity

To examine the consistency and reliability of the measurement results of latent variable data, a test was conducted on the measurement variable data used to characterize the latent variables, focusing on data reliability and validity. The test results are shown in Table 7.

4.3.1.1 | Reliability Analysis. The Cronbach's α coefficient was used as the indicator for reliability analysis. Based on the Nunnally standard, a coefficient α above 0.7 indicates a high level of internal consistency for each latent variable. From the α values obtained in the reliability analysis, it can be seen that the α values for each latent variable are all greater than 0.7, indicating a high internal consistency among the dimensions of the questionnaire, which is in line with the requirements for further research.

4.3.1.2 | Validity Analysis. The analysis of validity was conducted using Kaiser–Meyer–Olkin (KMO), Bartlett's sphericity test and average variance extracted (AVE). The analysis of various validity indicators shows that the KMO values are

TABLE 5 | Psychological latent variable measurement scale.

Latent variable	Measure indicator	Problem item
Subject norm	SN ₁	People around me recommended that I choose this travel mode/plan for travel.
	SN ₂	The news media recommended that I choose this travel mode/plan.
	SN ₃	I will consider others' suggestions on my travel mode/plan selection.
Behavioural attitude	BA ₁	For me, choosing this travel mode/plan is better than other modes/plans.
	BA ₂	Convenient decision-making—Using this travel mode/plan requires relatively less time and effort to view and compare service information before making a purchase decision.
	BA ₃	Transaction convenience—By utilizing this travel mode/plan, relatively less time and effort are spent in offline and online queuing and payment processes.
	BA ₄	After sales convenience—perceived to consume less time and effort when contacting the service provider again after completing core services.
	BA ₅	For me, choosing this travel mode/plan is very enjoyable.
Perceived behaviour control	PBC ₁	For me, choosing this travel mode/plan is very convenient and easy.
	PBC ₂	I have a high degree of freedom in choosing this travel mode/plan; it all depends on me.
	PBC ₃	For me, completing this trip was not difficult.
Behavioural intention	BI ₁	For me, I often choose this travel mode/plan.
	BI ₂	I will still choose this travel mode/plan for my next trip.
	BI ₃	I will recommend this travel mode/plan to people around me.
Travel habit	TH ₁	Choosing this mode of travel mode/plan has become my inherent habit.
	TH ₂	For my next trip, I would be inclined to choose this transportation mode/option.
	TH ₃	I usually do not consider choosing other modes of transportation/options for travel besides this one.
Perceived travel fairness	PF ₁	The fare of the chosen transportation mode/plan aligns with my psychological expectations under my personal economic conditions, and I am satisfied with its cost-performance ratio.
	PF ₂	The seating resources of the chosen travel mode/plan ensure that I can select my ideal seat.
	PF ₃	The chosen travel mode/plan requires a very short post-travel recovery time to alleviate fatigue.
	PF ₄	The chosen travel mode/plan offers a high level of safety performance.
	PF ₅	The chosen travel mode/plan demonstrates high punctuality and a strong capability to minimize spatiotemporal disparities.
	PF ₆	The service received is worth the time and effort I have invested.

all above 0.6; Bartlett's chi-square values are relatively large, with corresponding p values all less than 0.05, indicating the significance of Bartlett's sphericity test. Additionally, the AVE values for each latent variable are all greater than 0.5. Overall, this indicates that the convergent validity of the variables set is quite ideal.

4.3.2 | Distribution Characteristics of Measurement Indicators

The descending order of the mean and the distribution of the standard deviation of the measurement indicators corresponding to latent variables are shown in Figure 6.

TABLE 6 | Statistical analysis of socio-economic attributes of samples.

Category	Options	Sample (%)	Ticket data (%)
Gender	Male	47.89%	46.28%
	Female	52.11%	53.72%
Age	≤25	26.30%	25.67%
	≈26–35	55.58%	58.26%
	≈36–45	13.65%	12.93%
	≈46–55	3.27%	2.06%
	≈56–65	0.80%	1.01%
	>65	0.40%	0.07%
Educational level	Senior high school/Vocational school and below	3.72%	—
	Junior college/Vocational education	7.69%	—
	Undergraduate college	70.97%	—
	Graduate students and above	17.62%	—
Occupation	Student	18.11%	—
	Sales/Service personnel	6.95%	—
	Migrant workers	1.74%	—
	Employees of enterprises and institutions	59.31%	—
	Employees of government agencies	4.96%	—
	Individual entrepreneurs or freelancers	4.22%	—
	Retired personnel	0.74%	—
	Other	3.97%	—
Monthly income	≤3500 yuan	12.60%	—
	≈3501–6500 yuan	22.60%	—
	≈6501–9000 yuan	24.60%	—
	≈9001–15,000 yuan	26.05%	—
	≈15001–25,000 yuan	11.17%	—
	≥25001 yuan	2.98%	—

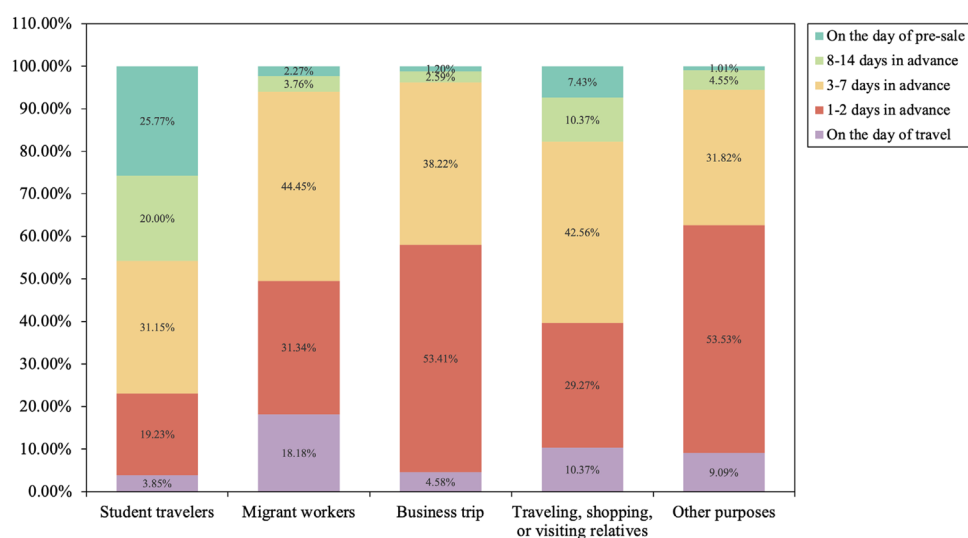


FIGURE 5 | Cross analysis of passenger travel purpose and number of days of early ticket purchase.

TABLE 7 | Reliability and validity analysis of latent variables.

Latent variable	Cronbach's α	KMO	Bartlett's	AVE
Subject norm (SN)	0.799	0.675	408.974	0.591
Behavioural attitude (BA)	0.880	0.874	995.515	0.599
Perceived behaviour control (PBC)	0.841	0.728	486.498	0.640
Behavioural intention (BI)	0.767	0.686	316.524	0.545
Travel habit (TH)	0.758	0.637	335.556	0.531
Perceived travel fairness (PF)	0.910	0.918	1462.184	0.630

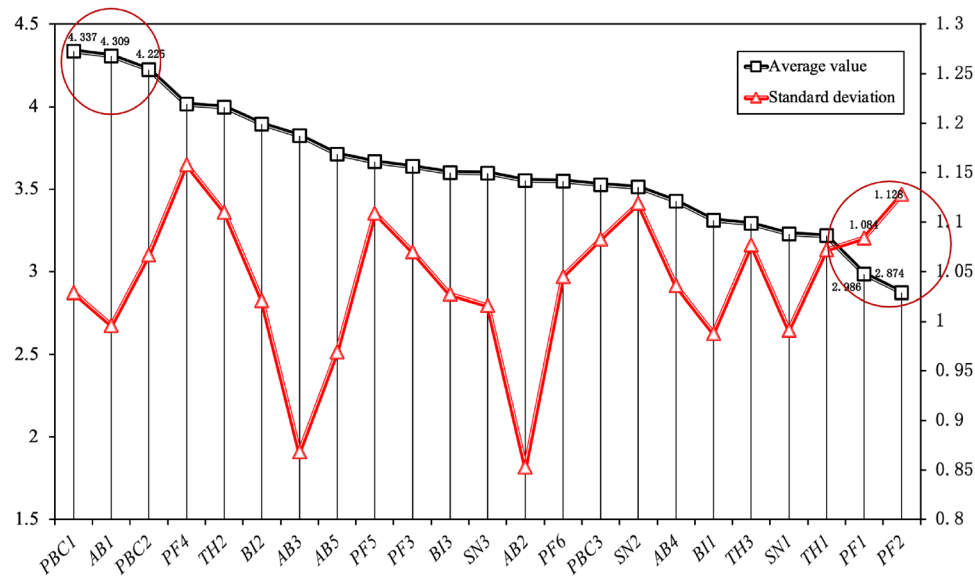


FIGURE 6 | Distribution characteristics of measurement indicator values.

As illustrated in Figure 6, the latent variable of perceived travel fairness shows that PF1 and PF2 received relatively low composite scores but exhibited significant standard deviations, indicating divergent opinions among respondents regarding the rationality of fare structures and the allocation of seating resources. This suggests that passengers' perception of fairness considerably influences their travel behaviour.

5 | Construction of Passenger Travel Behaviour Model Based on an Integrated Model

Given that passengers possess distinct individual and travel characteristics, it is essential to systematically account for the differences among these attributes when analysing travel decision-making processes. Doing so enhances the behavioural model's ability to represent, describe, and interpret travel choices.

Heterogeneity in passenger travel behaviour is mainly reflected in two aspects: preference heterogeneity and response heterogeneity. Preference heterogeneity refers to observable and unobservable differences in passengers' socio-economic attributes and psychological characteristics, while response heterogeneity captures the varying importance of travel option attributes across individuals, which may also be either observable or unobserv-

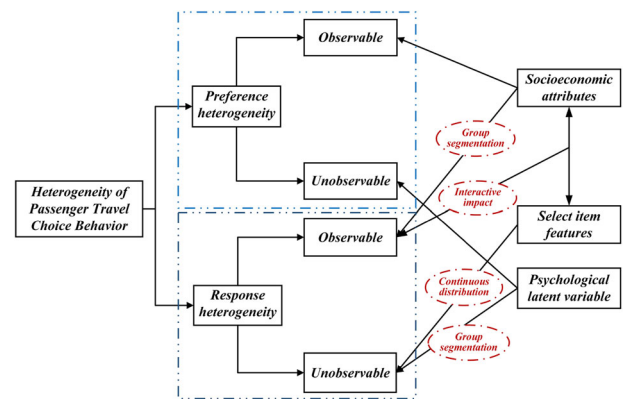


FIGURE 7 | Graphical representation of heterogeneity in passenger travel behaviour.

able. A graphical representation of passenger travel behaviour heterogeneity is provided in Figure 7.

It is evident that the behavioural heterogeneity among passenger groups significantly influences the optimization of railway service products and subsequent operational decision-making by transport enterprises. To further explore this issue, this section

employs advanced modelling techniques to develop a choice behaviour model for travel options. The aim is to provide an in-depth analysis of how both observed and latent variables within the decision-making framework affect the travel behaviour of heterogeneous passengers under dynamic pricing strategies.

5.1 | MIMIC Model

As a statistical model for establishing, estimating and testing causality, SEM can replace multiple regression, path analysis, covariance analysis and other methods to clearly analyse the role of single indicators on the overall and the relationship between single indicators. As a special structural equation model with a good representation effect to explain individual selection behaviour under complex perceptual state, SEM has special features in dealing with multiple indicators, considering the interaction of potential variables and introducing unequal measurement errors. In this study, the MIMIC model is applied to characterize passenger travel behaviour under dynamic pricing strategies.

The MIMIC model consists of a structural model and a measurement model. The structural model is used to characterize the relationships between latent variables of passengers, while the measurement model is used to characterize the relationships between latent variables and measurement indicators.

The distribution density function of the latent variables is defined as: $f_1(X_n^*|X_n; \alpha, \chi_\omega)$. The structural model is shown in Equation (2).

$$X_n^* = A(X_n; \alpha) + \omega_n, \omega_n \sim N(0, \chi_\omega) \quad (2)$$

where, X_n^* represents a latent variable; $A(\cdot)$ denotes a function, typically assumed to be linear; X_n stands for an explanatory variable; α is the parameter to be estimated; ω_n represents the random disturbance term; and χ_ω denotes the covariance. It should be noted that each latent variable corresponds to an Equation (2).

The distribution density function of the measurement indicator is defined as: $f_2(I_n|X_n, X_n^*; \beta, \chi_v)$. The measurement model is shown in Equation (3).

$$I_n = B(X_n, X_n^*; \beta) + v_n, v_n \sim N(0, \chi_v) \quad (3)$$

where, I_n represents the measurement variable; $B(\cdot)$ is a function similar to $A(\cdot)$ and is typically assumed to be a linear function as well; β is the parameter to be estimated; v_n represents the random disturbance term; and χ_v denotes the covariance. Each measurement indicator corresponds to an Equation (3).

In this regard, the MIMIC expression is defined as follows in this paper:

$$X_{iqn}^* = \sum_r \gamma_{iqr} Y_{irn} + \sum_q \varrho_{iqn} X_{iqn}^* + \omega_{iqn} \quad (4)$$

$$I_{ijn} = \sum_q \alpha_{iqj} X_{iqn}^* + \rho_{ijn} \quad (5)$$

where, Y_{irn} is a vector composed of socioeconomic attribute variables of passenger n that influence the latent variable X_{iqn}^* ; r represents the number of observable explanatory variables of socioeconomic attributes affecting X_{iqn}^* ; γ_{iqr} and ϱ_{iqn} are coefficient vectors to be determined; I_{ijn} is a vector composed of measurement indicators corresponding to X_{iqn}^* ; j denotes the number of measurement indicator variables for X_{iqn}^* ; α_{iqj} is a coefficient to be determined; ω_{iqn} and ρ_{ijn} are error terms.

Based on Sections 3 and 4, the structural model of passenger travel intention under a dynamic pricing strategy based on MIMIC can be summarized as shown in Figure 8.

In conclusion, based on the mathematical model combined with the theoretical model structure, the computational structure for solving the MIMIC model is derived. It is divided into two parts: the first part involves conducting confirmatory factor analysis on the structure of each X_{iqn}^* in the high-speed rail travel intention model under dynamic pricing strategy, and using AMOS software to construct its path analysis as shown in Figure 9

In the MIMIC model analysis, when the measurement unit of X_{iqn}^* cannot be determined, uncertainty should be removed through standardization during model estimation. Typically, the path coefficient of a specific indicator variable β for the latent variable is set to 1.

The second part involves analysing the quasi-regression relationship between Y_{irn} and X_{iqn}^* . Taking the latent variable PF, denoted as X_{PF}^* as an example:

$$X_{PFn}^* = \alpha_{PF1} Y_{1n} + \alpha_{PF2} Y_{2n} + \dots + \alpha_{PF6} Y_{6n} + \omega_{PFn} \quad (6)$$

In the equation, α_{PFr} is the parameter to be estimated; Y_{1n} to Y_{6n} represent observable explanatory variables such as gender, age, education level and other socio-economic attributes.

5.2 | LCM Framework

Market segmentation based on travel behaviour data assists railway operators in precisely formulating marketing strategies, retaining frequent passengers and stabilizing those who may switch to other modes of transportation. This section examines the response heterogeneity exhibited by high-speed rail passengers in their travel mode/option choices under dynamic pricing strategies by establishing LCM. The LCM assumes that the study population can be categorized into a finite number of groups with homogeneous preferences according to a specific probabilistic allocation model. The framework of the model is summarized in Figure 10.

5.2.1 | Probability Parameterization and Model Construction

1. Latent class probability- P_t^X and conditional probability- $P_t^{(A,B,C,\dots)/X}$

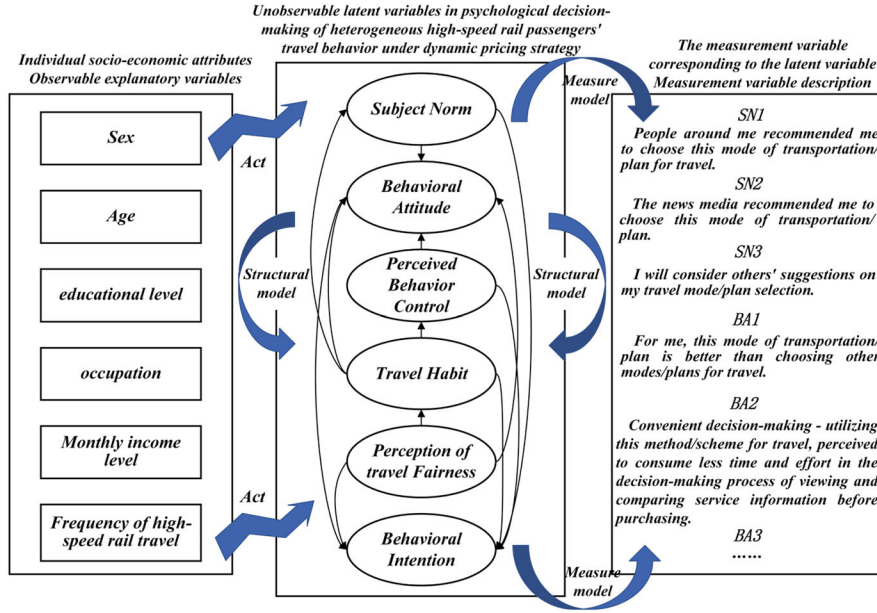


FIGURE 8 | The structure of passenger travel behaviour intention model under dynamic pricing strategy based on MIMIC.

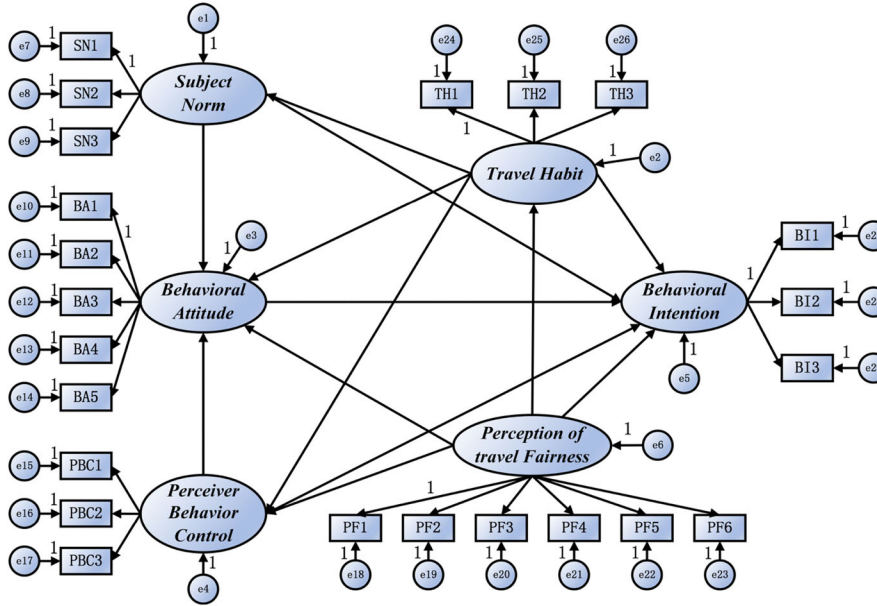


FIGURE 9 | The path analysis diagram of passenger travel behaviour intention model under dynamic pricing strategy.

When classifying passengers, latent class probabilities represent the probability of a passenger being assigned to class t , used to characterize the parameters of the latent variable X . That is, $P_t^X, t = 1, 2, \dots, T$.

Assuming there are exogenous variables $\{A, B, C\}$ each containing multiple levels, where $A = \{a_1, a_2\}$, $B = \{b_1, b_2, b_3\}$, $C = \{c_1, c_2\}$. The conditional probability refers to the probability that, within the t th category of each X , a randomly selected passenger sample makes a specific choice at a certain level of the exogenous variables, denoted as $P_t^{(A,B,C,\dots)/X}$.

Since, under a certain T , any passenger must make a unique choice in different levels of an explicit variable. Therefore, the

sum of conditional probabilities of all levels of an explicit variable is 1, which is shown in Equation (7).

$$\sum_{a_1=1}^A P_t^{A=a_1/X} = \sum_{b_1=1}^B P_t^{B=b_1/X} = \sum_{c_1=1}^C P_t^{C=c_1/X} = 1 \quad (7)$$

2. Model representation

The LCM is a joint probability expression. For a specific passenger, when they belong to the levels of exogenous variables $\{A, B, C\}$ as $\{a_1, b_2, c_3\}$, respectively, the joint probability expression is given by $P_t^{A=a_1, B=b_2, C=c_3/X}$. The LCM model representation is

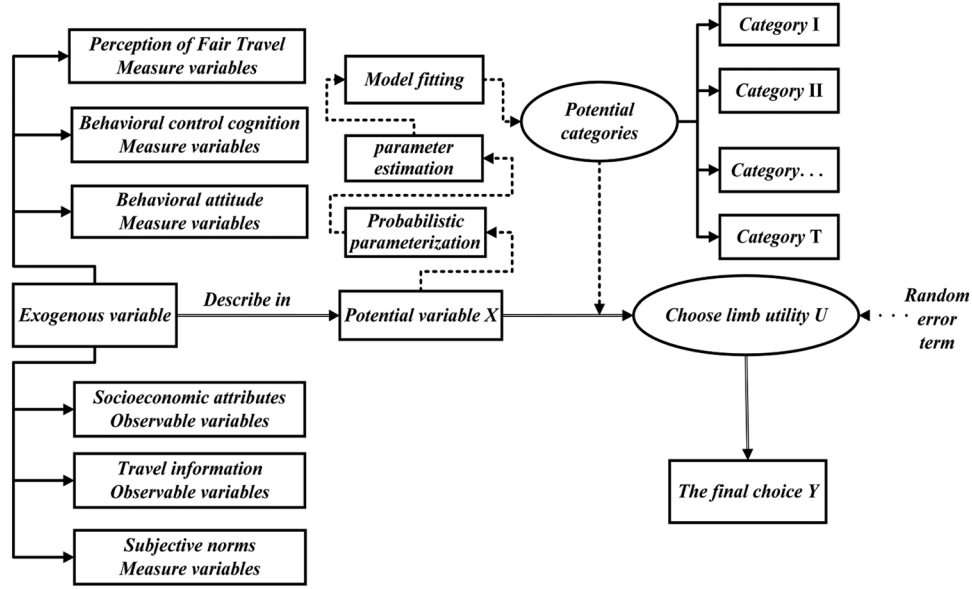


FIGURE 10 | LCM model framework.

shown in Equations (8) to (10).

$$P_t^{\frac{A=a_2, B=b_3, C=c_1}{X}} = \sum_{t=1}^T P_t^X \prod_{n=1}^3 P_t^{\frac{A,B,C=\{a_n, b_n, c_n\}}{X}} \quad (8)$$

$$\text{s.t., } \sum_{t=1}^T P_t^X = 1 \quad (9)$$

$$\sum_{a_1=1}^A P_t^{\frac{A=a_1}{X}} = \sum_{b_1=1}^B P_t^{\frac{B=b_1}{X}} = \sum_{c_1=1}^C P_t^{\frac{C=c_1}{X}} = 1 \quad (10)$$

5.2.2 | Parameter Estimation and Model Fitting

The maximum likelihood estimate (MLE) serves as the primary method for estimating the parameters of the LCM. In this study, we employ the expectation-maximization (EM) algorithm for an iterative solution. In the LCM, it is assumed that the latent categorical variable X has T latent classes. The constructed likelihood function is shown in Equation (11):

$$\hat{P}_t^{\frac{(A,B,C)}{X}} = \prod_{n=1}^3 \hat{P}_t^X \hat{P}_t^{\frac{A,B,C=\{a_n, b_n, c_n\}}{X}} \quad (11)$$

where, $\hat{P}_t^{\frac{(A,B,C)}{X}}$ represents the probability of the observed variables $\{A, B, C\}$ given the latent variable X occurring in the t th class; $\hat{P}_t^X$ denotes the estimated probability of the latent class; and $\hat{P}_t^{\frac{A,B,C=\{a_n, b_n, c_n\}}{X}}$ stands for the maximum likelihood conditional probability estimate of the observed variables. To test the applicability of the model, the probability estimates of each explicit variable under the potential categories are added and summed to obtain the maximum likelihood estimator of the joint probability,

as shown in Equation (12).

$$\hat{P}_t^X = \sum_{t=1}^T \hat{P}_t^{\frac{(A,B,C)}{X}} = \sum_{t=1}^T \prod_{n=1}^3 \hat{P}_t^X \hat{P}_t^{\frac{A,B,C=\{a_n, b_n, c_n\}}{X}} \quad (12)$$

where, $\hat{P}_t^X$ is expressed as the probability that the explicit variable $\{A, B, C\}$ appears in the t th category. Transform the two equations to obtain the maximum likelihood estimation probability of each explicit variable in each potential category and at different levels, as shown in Equation (13).

$$\hat{P}_t^{\frac{(A,B,C)}{X}} = \frac{\hat{P}_t^{\frac{(A,B,C)}{X}}}{\hat{P}_t^X} = \frac{\prod_{n=1}^3 \hat{P}_t^X \hat{P}_t^{\frac{A,B,C=\{a_n, b_n, c_n\}}{X}}}{\sum_{t=1}^T \prod_{n=1}^3 \hat{P}_t^X \hat{P}_t^{\frac{A,B,C=\{a_n, b_n, c_n\}}{X}}} \quad (13)$$

where, $\hat{X}$ represents the sum of probabilities of latent classes, and $\hat{P}_t^{\frac{(A,B,C)}{X}}$ denotes the proportion of each level of observed variables in a particular latent class, serving as the relative importance of each observed variable in that latent class.

5.2.3 | Latent Classification

Latent class analysis aims to categorize the observed data of all passengers into appropriate latent classes. Assuming a Logit model form for the classification model, the probability of passenger- i being assigned to latent class- t in this paper is shown in Equation (14).

$$P_i(t) = \frac{\exp(\lambda_t + \alpha_t^T Z_i + \beta_t^T W_i)}{\sum_{t=1}^T \exp(\lambda_t + \alpha_t^T Z_i + \beta_t^T W_i)} \quad (14)$$

In the equation, λ_t represents the constant term for the latent class t ; α_t denotes the estimated parameters corresponding to the individual attributes variable Z_i in class t ; β_t represents the estimated parameters corresponding to the travel-related variables W_i in class t , including attributes such as passengers'

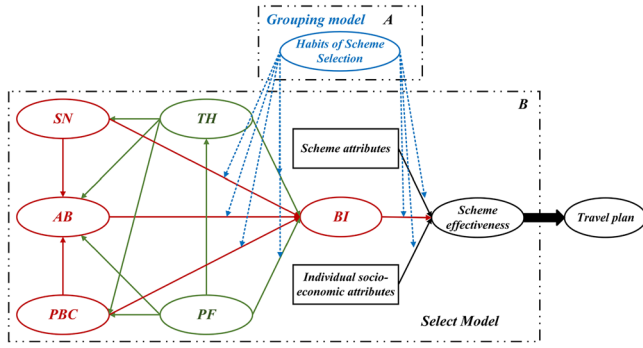


FIGURE 11 | MIMIC-LCM integrated model framework.

travel habits, perceived travel fairness, and behavioural intentions towards travel options.

5.3 | MIMIC-LCM Integrated Model

This paper creatively combines TPB theory with LCM to construct a MIMIC-LCM integrated model, which comprehensively considers and explains the heterogeneity of passengers from the perspective of potential variables and potential categories. The choice model-MIMIC model is selected to analyse preference heterogeneity from the perspective of psychological latent variables and reveal the role of this heterogeneity in the decision-making process of travel mode choices. The grouping model LCM is used to analyse heterogeneity from the perspective of group segmentation, while also explaining the differences in the effects of response heterogeneity on travel decision-making behaviour for the same observed variable across groups.

In summary, the MIMIC-LCM integrated model integrates preference heterogeneity and response heterogeneity to further enrich the characteristics of heterogeneous passenger choice behaviour. The model framework is illustrated in Figure 11.

In the Figure 11, the small dashed box represents the grouping model, while the large dashed box represents the choice model. The black section represents the mechanism of traditional travel mode choice models. The red section represents the various latent variables and their interactions in the TPB. The green section marks the expanded latent variables. The blue section involves the grouping variables of the LCM model, namely travel mode choice habits and their influence on the choice model.

Overall, the integrated MIMIC-LCM model has excellent characteristics for behavioural explanation, including diverse types of influencing factors, clear variable relationships, and a comprehensive research framework.

6 | Behaviour Analysis Based on MIMIC-LCM Integrated Model

6.1 | Parameter Estimation and Result Analysis of Grouping Model

The classification was based on the statistical results of high-speed rail passengers' choices under 92 sets from the 'High-Speed

Rail Passenger Travel Choice Behaviour Survey' questionnaire. These scenarios incorporated six attributes: departure time, travel time, train occupancy period, service level, ticket price and advance purchase time. The levels of each attribute were processed as shown in Table 8.

6.1.1 | Model Fitting Results

To determine the optimal number of classes, the model fit was evaluated using four statistical indicators: Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), the Pearson χ^2 statistic, and the likelihood ratio chi-square (G^2). Lower values of these indices indicate a better model fit. The test results are summarized in Table 9.

As shown in Table 9, as the number of classes increases, the AIC, BIC and Pearson χ^2 values reach their minimum when the number of clusters is 4, and begin to rise again from cluster = 5 onward. The G^2 statistic reaches its minimum at cluster = 6, though the values at cluster = 4 and 5 show little practical difference. Considering the operational characteristics of railway passenger transport, the final number of classes should yield actionable segments for strategy implementation, and the classification dimensions along with their subdivisions should remain parsimonious. Based on these considerations, a latent class count of 4 was selected as the ideal number of passenger segments.

6.1.2 | Model Parameter Estimation

For cluster = 4, the EM algorithm is used to estimate the latent class probabilities and conditional probabilities of the model. The results are shown in Table 10.

As indicated in Table 10, Class 1 passengers account for 24.37% of the sample. This group primarily values time-related factors in their travel decision-making, showing a strong preference for options with shorter travel durations. Most of these passengers book their tickets within 1 week before departure, with a notably high proportion of 42% purchasing tickets 1–2 days in advance. They exhibit relatively low sensitivity to changes in the economic and comfort attributes of travel options. This profile aligns well with business travellers on official trips, who can accept a certain degree of fare flexibility within a reasonable range. Compared to other passenger classes, a larger proportion of this group opted for dynamically priced options that depart later than the time slots available under static pricing. Based on these characteristics, this segment is defined as 'Time-Sensitive Passengers'.

Class 2 passengers account for 20.03% of the sample. These passengers prioritize travel cost when selecting trip options, showing a clear preference for lower-priced alternatives. They are willing to tolerate lower train speed categories and reduced comfort in exchange for economy, and are relatively indifferent to departure times. To increase the likelihood of securing their desired options, most purchase tickets within the first week of the advance sale period, with a notably high rate of 32% choosing to buy tickets on the first day tickets become available. Based on these behavioural traits, this group is defined as 'Cost-Sensitive Passengers'.

TABLE 8 | Attribute level processing.

Attribute name	External variable	Classification level	Note information (ascending order)
Departure time	DT	7	(1) 6 h in advance; (2) 4 h in advance; (3) 2 h in advance; (4) original time; (5) delay by 2 h; (6) delay by 4 h; (7) delay by 6 h
Travel time	TT	3	(1) 4.5 h; (2) 5.5 h; (3) 6.5 h
Train occupancy period	TP	2	(1) High occupancy period; (2) Low occupancy period
Service level	TSL	5	(1) Especially comfortable; (2) Relatively comfortable; (3) Generally comfortable; (4) Relatively uncomfortable; (5) Extremely uncomfortable
Ticket price	TP	5	(1) Original price 80%; (2) Original price 90%; (3) Original ticket price; (4) Original price 110%; (5) Original price 120%
Advance ticket purchase time	PT	5	(1) On the day of departure; (2) 1–2 days in advance; (3) 3–7 days in advance; (4) 8–14 days in advance; (5) On the day of pre-sale

TABLE 9 | Significance test criteria for estimated parameters.

Cluster	AIC	BIC	Pearson χ^2	G ²
2	6618.371	6757.513	1627.417	1372.103
3	6132.707	6397.833	1359.223	1107.613
4	5792.163	5917.649	1031.706	924.379
5	5837.227	5987.412	1103.751	893.711
6	5896.718	6037.281	1187.428	651.431

Class 3 constitutes the largest segment, representing 41.27% of passengers. These individuals exhibit moderate acceptance of fare increases and a relative preference for earlier departures and higher-speed trains, yet show no strong, distinct preference across attribute levels. This group is accordingly defined as ‘Scenario-Dependent Passengers’.

Class 4 passengers account for 14.33% of the sample. These passengers place high importance on travel efficiency and comfort, generally showing little concern for fare increases. They tend to purchase tickets within the week prior to departure. Compared to the departure times available under static pricing, they prefer dynamically priced options departing at adjacent times, and avoid options with excessively early departures or late arrivals. This segment is therefore classified as ‘Experience-Oriented Passengers’.

Using Class3 as a reference, the factors of gender, age, education level, occupation, and monthly income level are selected as the horizontal factors to analyse the relationship between the individual socio-economic attributes of passengers and the latent categories. The results are shown in Table 11.

As shown in Table 11, passengers’ monthly income level exerts the most significant influence across the three latent classes. The negative coefficient for Class 2 (cost-sensitive passengers) indicates that higher-income passengers are more sensitive to travel time and comfort rather than cost, and are more likely to belong to Class 1 or Class 4.

Gender significantly affects Class 2 and Class 4. The positive coefficient for Class 2 suggests that female passengers are more cost-sensitive and more likely to fall into this category. In contrast, the negative coefficient for Class 4 implies that male passengers are more sensitive to travel experience and are more likely to be classified as experience-oriented passengers. Age has a significant impact on all three latent classes. The negative coefficients for Class 1 and Class 2 indicate that older passengers are less likely to be time-sensitive or cost-sensitive. The positive coefficient for Class 4 suggests that older passengers pay more attention to travel comfort and are more likely to be classified as experience-oriented.

Education level also shows a significant influence across all classes. The negative coefficient for Class 2 and positive coefficients for Class 1 and Class 4 imply that highly educated passengers are more sensitive to travel time and comfort, making them more likely to belong to Class 1 or Class 4, while less educated passengers show stronger cost sensitivity and are more likely to be assigned to Class 2. Similarly, occupation significantly affects all three classes. The negative coefficient for Class 2 and positive coefficients for Class 1 and Class 4 indicate that passengers with higher professional status and retired individuals are more sensitive to travel time and comfort, and thus more likely to belong to Class 1 or Class 4. In contrast, passengers with lower occupational status, such as students, exhibit stronger cost sensitivity and are more likely to be categorized as cost-sensitive passengers.

6.2 | Parameter Estimation and Result Analysis of Select Model

Building upon the classification results from the segment-specific models, a discrete choice model was integrated to interpret the behavioural characteristics of each passenger subclass under complex perceptual states. The SEM is constructed for each category of passengers to strengthen the model. Through modification, the path coefficient and path relationship between potential variables of the intention model are shown in Figure 12 below. Here, ‘**’ and ‘***’ denote statistical significance at the

TABLE 10 | Results of conditional probability and potential category probability.

Attribute		Class 1	Class 2	Class 3	Class 4
DT	1	0.022	0.132	0.072	0.024
	2	0.105	0.188	0.186	0.107
	3	0.168	0.219	0.215	0.221
	4	0.329	0.247	0.224	0.317
	5	0.210	0.103	0.192	0.218
	6	0.114	0.097	0.094	0.092
	7	0.052	0.014	0.017	0.021
TT	1	0.718	0.176	0.346	0.531
	2	0.237	0.457	0.437	0.327
	3	0.045	0.367	0.217	0.142
TP	1	0.513	0.542	0.497	0.383
	2	0.487	0.458	0.503	0.617
TSL	1	0.103	0.097	0.081	0.242
	2	0.117	0.103	0.238	0.373
	3	0.461	0.496	0.396	0.237
	4	0.263	0.243	0.217	0.096
	5	0.056	0.061	0.068	0.052
TP	1	0.003	0.573	0.245	0.012
	2	0.094	0.307	0.364	0.103
	3	0.292	0.117	0.173	0.225
	4	0.398	0.002	0.121	0.423
	5	0.213	0.001	0.097	0.237
PT	1	0.328	0.003	0.051	0.128
	2	0.424	0.201	0.234	0.263
	3	0.227	0.237	0.242	0.330
	4	0.019	0.241	0.237	0.216
	5	0.002	0.318	0.236	0.069
Potential category probability		0.24373	0.20031	0.41269	0.14327

0.05 and 0.01 levels, respectively. Solid red lines indicate that the original hypothesized path is supported, with the corresponding path coefficient provided. Black lines marked with ‘x’ indicate that the hypothesized relationship was not supported, and the path has been removed from the model.

Based on the specific connotations of the latent and measured variables as described in Tables 3 and 5, an internal horizontal analysis was conducted for each passenger category, with the results as follows:

1. The path relationships among the latent variables for Class 1 (time-sensitive passengers), as shown in Figure 12a, indicate that: (a) Since both BA and PF involve time-related attributes, they exert a strong positive and significant influence on BI; (b) PF has a strong positive and significant impact on both BA and TH, indicating that a stronger sense of fairness under dynamic pricing strategies makes time-sensitive passengers more inclined to habitually choose high-speed

rail and enhances their overall evaluation of it; (c) As most passengers in this category are business travellers, the positive and significant influence of SN and TH on BA suggests that recommendations from colleagues and the frequency of past high-speed rail usage for business trips positively shape their overall perception of high-speed rail travel.

Based on the path relationships between the latent variables and their corresponding measured indicators, the following observations can be made: (1) Indicators PF₃ and PF₅ show strong positive and significant effects on PF, while PF₁ has a relatively weaker influence. This suggests that passengers in this category place greater importance on the speed grade and punctuality of dynamically priced travel options rather than economic factors, which is consistent with the classification results of the segment-specific model. (2) Indicator TH₂ exhibits the strongest positive and significant impact on TH, indicating that the high accessibility and reliability of high-speed rail make it the preferred mode of travel for business trips among this

TABLE 11 | Personal socio-economic attributes and potential category estimation test.

Clusters	Character	Coefficient	Std. Error	$P > Z $	95% Conf. Interval []	
Class 1	Gender	0.263179	0.317458	0.300	-0.237197	0.802017
	Age	-0.320792*	0.112705	0.013	-0.566019	0.052721
	Education	0.185462*	0.106591	0.098	-0.025978	0.341725
	Occupation	0.253117*	0.230167	0.017	0.030217	0.481678
	Income	0.397121***	0.071845	0.000	0.234127	0.572203
	cons_	-1.872654	0.630728	0.127	-3.426187	0.532771
Class 2	Gender	0.762147**	0.231277	0.003	0.279814	1.270122
	Age	-0.283981**	0.127466	0.007	-0.461242	-0.081369
	Education	-0.255712*	0.142078	0.028	-0.451926	-0.018807
	Occupation	-0.273618*	0.181637	0.018	-0.465101	-0.050219
	Income	-0.357916***	0.082601	0.000	-0.522670	-0.162958
	cons_	0.972318	0.710322	0.138	-0.331971	2.321017
Class 4	Gender	-0.421766**	0.103473	0.001	-0.752319	-0.130118
	Age	0.362197*	0.142764	0.036	0.042317	0.637145
	Education	0.142213*	0.237736	0.020	-0.137465	0.327019
	Occupation	0.312793*	0.181467	0.023	0.017019	0.572633
	Income	0.381127***	0.087419	0.000	0.047183	0.612790
	cons_	0.892614	0.607138	0.207	-0.352970	2.276470

Note: *, ** and *** represent significant differences at the levels of 0.1, 0.05 and 0.01, respectively.

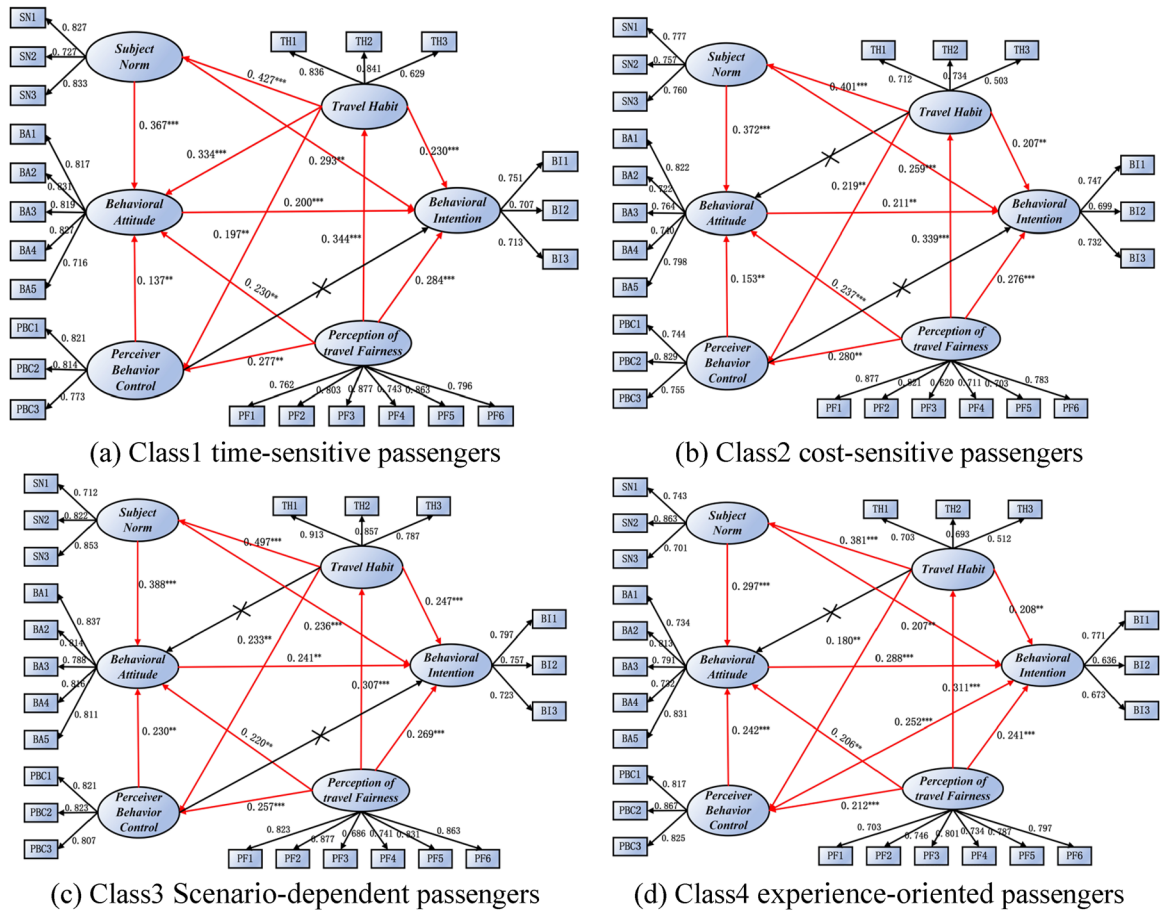


FIGURE 12 | Path relationship between latent variables in the intention model.

passenger group. (3) Indicators BA_2 and BA_4 significantly and positively influence BA, implying that these passengers value decision-making convenience and after-sales service flexibility, aligning with the typical needs of business travellers.

- As shown in the path relationships among latent variables for Class 2 (cost-sensitive passengers) in Figure 12b: (1) Since PF involves economic attributes, it has a strong positive and significant effect on BI, indicating that this passenger group is highly concerned with the economic aspects of dynamically priced travel options. (2) PF also shows strong positive effects on PBC and TH, suggesting that enhanced fairness perceptions under dynamic pricing strategies reinforce their habitual use of high-speed rail and increase their perceived freedom of choice.

Regarding the relationships between latent variables and their measured indicators: (1) Indicator PF_1 has the most significant positive effect on PF, indicating that this group is primarily influenced by the economic characteristics of dynamic pricing offers, with less emphasis on speed or punctuality—consistent with the segment model analysis. (2) Indicators BA_1 and BA_5 significantly and positively affect BA, showing that lower travel costs lead to higher overall satisfaction, and through the positive effect of BA on BI, higher satisfaction strengthens willingness to pay. (3) The influence of indicator TH_3 on TH is noticeably weaker compared to the other two indicators. Due to the price-sensitive nature of this group, their choice of travel option is highly susceptible to current pricing. If the actual price deviates significantly from their psychological expectation, they are likely to switch to alternative transport modes.

- As illustrated by the path relationships among the latent variables for Class 3 (scenario-dependent passengers) in Figure 12c: (1) SN, BA, TH and PF all exhibit positive and significant effects on BI, reflecting this group's strong reliance on high-speed rail as a travel mode. (2) TH has a significant positive influence on SN, indicating that passengers who habitually choose high-speed rail are more likely to recommend it to others and have their advice accepted under dynamic pricing settings. (3) PBC shows a significant positive effect on BA, suggesting that the easier and more controllable passengers perceive the travel choice to be, the more positively they evaluate high-speed rail under this strategy.

Regarding the relationships between latent variables and their measured indicators: (1) Indicators BI_1 and BI_2 have notable positive effects on BI, implying strong demand and dependency on high-speed rail among this group. (2) Indicator PF_2 exerts the most significant positive effect on PF, indicating that these passengers prioritize the availability of seating resources that meet their travel needs. In contrast, PF_1 , PF_3 and PF_5 show weaker effects, suggesting lower concern for economic factors, travel time efficiency, or speed level—consistent with the segment-based model analysis. (3) For the latent variable SN, indicator SN_3 has a more pronounced positive effect compared to others, implying that travel choices under dynamic pricing are primarily based on personal preference rather than external information sources.

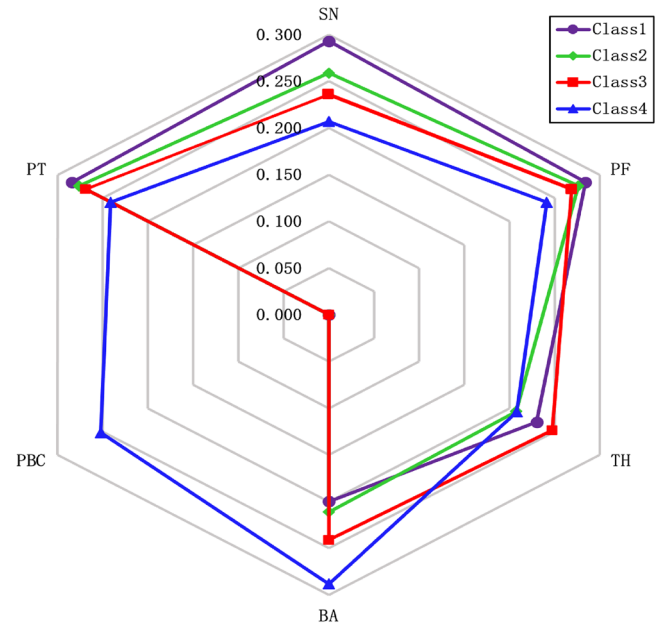


FIGURE 13 | Distribution of significant impact of latent variables of various types of passengers on BI.

- As observed in the path relationships among latent variables for Class 4 (experience-oriented passengers) in Figure 12d: The latent variables BA and PBC both show significant positive effects on BI, indicating that this group's intention to choose high-speed rail under dynamic pricing is mainly influenced by their overall evaluation of the option and their sense of control. This aligns with their high demand for travel quality and reflects a shift from 'wanting' to 'experiencing' comfort and efficiency.

In terms of the relationships between latent variables and their measured indicators: (1) For PF, indicator PF_3 has the most significant positive effect, while PF_1 has a minor influence. This indicates that these passengers are more concerned with the comfort and speed of dynamically priced options rather than economic factors. (2) Regarding TH, indicator TH_1 contributes most significantly to the latent variable, suggesting that their habitual choice of high-speed rail under dynamic pricing is largely influenced by their established behavioural patterns.

To illustrate the influence characteristics of latent variables across different passenger groups, a longitudinal analysis was conducted for each latent variable and its sub-measurement indicators under the perceived travel fairness construct. The results are shown in Figures 13 and 14.

- The positive influence of SN on BI is relatively weaker for Class 4 passengers compared to the other three categories. This can be attributed to their stronger consumption capacity and higher decision threshold, resulting in a lower likelihood of being influenced by recommendations from others when making travel choices under dynamic pricing strategies. Class 3 passengers, due to their context-dependent reliance on high-speed rail, are less sensitive to changes in product or pricing models. In contrast, Class 1 and Class 2 passengers, who exhibit distinct concerns about train attributes (time and cost, respectively), pay greater

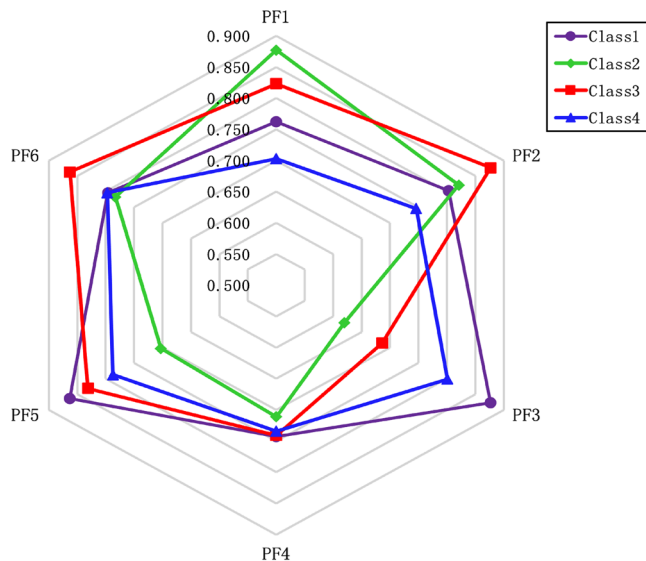


FIGURE 14 | Distribution of the impact of PF sub indicators on various types of passengers.

attention to external information sources, leading to a more significant positive impact of SN on their behavioural intentions.

(2) PF demonstrates a notably significant positive influence on BI across all four passenger categories compared to other latent variables. Specifically, Class 1 passengers prioritize time-related fairness, Class 2 emphasizes price fairness, Class 3 focuses on opportunity fairness and Class 4 values outcome-related fairness. These findings highlight the need for railway operators to clarify the relationship between travel experience and passenger satisfaction, and to establish objective criteria for evaluating travel fairness. Doing so will help enhance passengers' willingness to choose high-speed rail under dynamic pricing, improve their sense of fulfilment, and contribute to the development of a sustainable and equitable transportation ecosystem.

6.3 | Model Discussion

In summary, integrated model construction constitutes an iterative and progressively refined process. The choice model, grounded in the relationships between latent variables across passenger segments and their associated measurement variables, enhances the classification outcomes of the segmentation model. This approach verifies that passengers with distinct psychological profiles exhibit significantly different behavioural responses to the same perceptual factors when making travel decisions under dynamic pricing strategies. While prior research acknowledges that decision-making mechanisms vary across travel alternatives, existing studies often simplistically assume that selecting a given alternative reflects habitual behaviour, whereas choosing an option with strongly preferred attributes constitutes an intentional, rational decision. This dichotomy overlooks the nuanced interplay between psychological predispositions and contextual pricing stimuli in modal choice formation.

The MIMIC-LCM integrated model proposed in this study demonstrates that this is not a universal conclusion; rather, it

depends on the psychological characteristics of the traveller. For instance, when passengers show strong dependence on time, cost, or comfort, their choice of an option with high weight on the corresponding attribute is more likely a habitual and automatic behaviour shaped by specific contexts. In contrast, for context-dependent passengers, the selection of a highly weighted attribute option is more likely the outcome of a deliberate and rational process. Passenger groups with different psychological traits exhibit heterogeneity in how socioeconomic attributes and option characteristics influence their choices. Moreover, within the extended TPB framework, the effects of various perceptual factors on behavioural intention—and ultimately on mode choice—also differ significantly across groups. The latent variable perceived fairness consistently exerts a more substantial positive influence on travel intention compared to other latent variables. Additionally, the nature of the choice process differs among latent passenger classes: Time-sensitive, cost-sensitive and experience-oriented passengers exhibit inherent behavioural patterns, whereas scenario-dependent passengers engage in more rational and deliberative decision-making.

Existing behavioural analysis models exhibit limited applicability—while flexible in expression, they fail to clearly identify latent passenger segments. Moreover, the absence of a comprehensive theoretical framework integrating the psychological processes underlying HSR travel choices under dynamic pricing strategies has hindered the exploration of interrelationships among psychological factors. This gap has led to insufficient clarity among enterprise managers regarding the coupling mechanisms between fare structures and travel behaviour. The integrated model proposed in this study provides a dual perspective—latent variables and latent class analysis—to deeply dissect heterogeneous passenger choice behaviour, revealing behavioural traits and enabling precise passenger segmentation. Consequently, transportation demand management strategies should incorporate differentiated approaches based on an understanding of passengers' psychological needs and their variations. By aligning policies and measures with specific behavioural determinants and target groups, managers can ensure maximum effectiveness in implementation.

7 | Conclusion

This paper integrates social psychology and behavioural science to investigate the behavioural mechanisms underlying high-speed rail passengers' choice behaviour under dynamic pricing strategies, taking into account passenger heterogeneity. By extending the traditional TPB through the introduction of two latent variables—'travel habit' and 'perceived travel fairness'—a psychological decision-making framework was developed to quantify travel behaviour under dynamic pricing while accounting for passenger heterogeneity. A survey was conducted to examine the choice behaviour of heterogeneous passenger groups under dynamic pricing conditions. A D-optimal experimental design was employed to create realistic dynamic pricing service scenarios, enabling accurate collection of passenger choice data. To thoroughly examine the impact of both preference heterogeneity and response heterogeneity on travel choices, an integrated MIMIC-LCM model was constructed combining

structural equation modelling and latent class methods. This model provides a dual perspective—through latent variables and latent classes—for in-depth analysis of heterogeneous passenger behaviour, revealing behavioural traits and facilitating passenger segmentation. Based on the survey data, model calibration results demonstrate that the proposed integrated model achieves a good fit. Passengers were segmented into four distinct categories: time-sensitive, cost-sensitive, scenario-dependent, and experience-oriented. The model systematically captures each group's choice habits and preference characteristics, offering a more precise depiction and analysis of the behavioural patterns and influencing mechanisms of heterogeneous passengers under dynamic pricing. This enriches the theoretical foundation of passenger travel choice modelling and establishes a theoretical basis, and provides key parameter inputs for subsequent pricing research.

Future studies could incorporate additional extended variables—such as ‘environmental concern’ and ‘transfer convenience’—by integrating practical considerations or other behavioural theories to further refine the decision-making framework for heterogeneous high-speed rail passengers and improve model performance. Moreover, the findings indicate that the perceived fairness of pricing strategies plays a significant role in passengers’ travel choices. Given the strong public welfare nature of high-speed rail services, subsequent research on dynamic pricing should incorporate passengers’ perception of price fairness into tariff strategy design for segmented passenger markets. This approach would help achieve a dynamic balance between railway revenue and the public value of transportation services.

Author Contributions

Siye Guo: conceptualization, methodology, software, writing – original draft, writing – review and editing, visualization. **Yun Jing:** validation, writing – review and editing, supervision, funding acquisition. **Siyuan Wang:** writing – review and editing, validation. **Hengzhi Liu:** investigation, data curation, visualization. **Yiao Liu:** formal analysis, visualization.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data included in this study are available upon request by contacting the corresponding author.

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